Machine Learning(CS3104)

K K Singh,
Dept of CSE, RGUKT Nuzvid

Outcomes

Students will be able to

- Understand the basic theory underlying machine learning.

What is machine learning

- A branch of **artificial intelligence**, concerned with the design and development of algorithms that allow computers to evolve behaviors based on empirical data.
- Machine Learning is to
  - Develop algorithms that
  - improve their performance (P)
  - at some task (T)
  - with experience (Dataset: D)

- So there are the three components so you need a **class of tasks** you need a **performance measure** and you need **some well-defined experience** .

What is machine learning

Example

- Teach a kid to recognize an object (say : Pen) in initial stage of life:
- Task (T): Classify object: whether it is pen or not
- Experience (Data set D): Sample images (shape size, color) of different types of pen

Construct a machine learning (Classification) model in the kid brain.

- A Performance m

Source: https://www.google.com/imgres?imgurl=https%3A%2F%2Fpreviews.123rf.com%2Fimages%2Ffamveldman%2Ffamveldman1506%2Ffamveldman150600014%2F40479215-child-with-draw-and-paint-supplies-kids-happy-to-go-back-to-school-preschool-kid-learning-and-studyi.jpg&imgrefurl=https%3A%2F%2Fwww.123rf.com%2Fphoto_40479215_child-with-draw-and-paint-supplies-kids-happy-to-go-back-to-school-preschool-kid-learning-and-studyi.html&docid=KlhwCGZe_r7DsM&tbnid=hNvBs58t6Y4yyM%3A&vet=10ahUKEwi-3bS4p5ndAhUaWX0KHXplCYsQMwi8Aig3MDc..i&w=1300&h=866&bih=662&biw=1360&q=kid%20learning&ved=0ahUKEwi-3bS4p5ndAhUaWX0KHXplCYsQMwi8Aig3MDc&iact=mrc&uact=8

Machine Learning (Cont…)

Contents:

- ML Paradigms
- Why Learn?
- Application

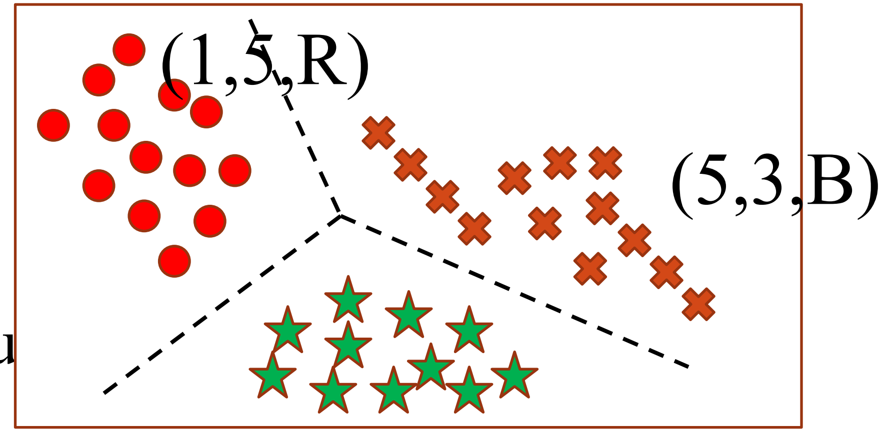
ML Paradigms

1. Supervised Learning

--Learn an input output map

Classification: categorical outpu

Regression: Continuous output.

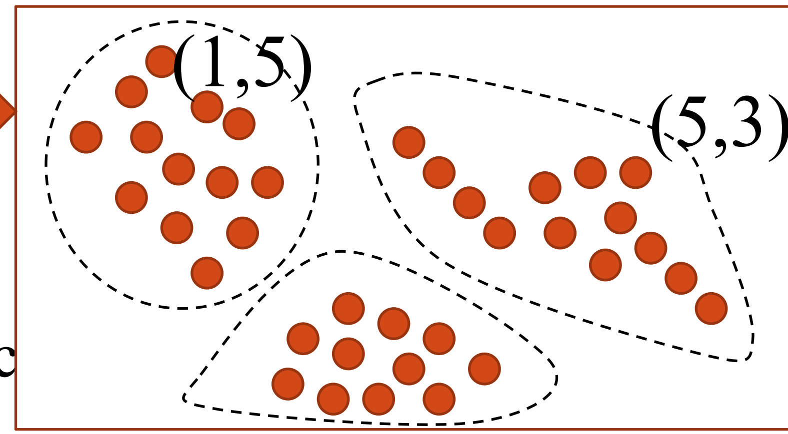


2. Unsupervised Learning

--Discover Pattern in the data

Clustering: cohesive grouping

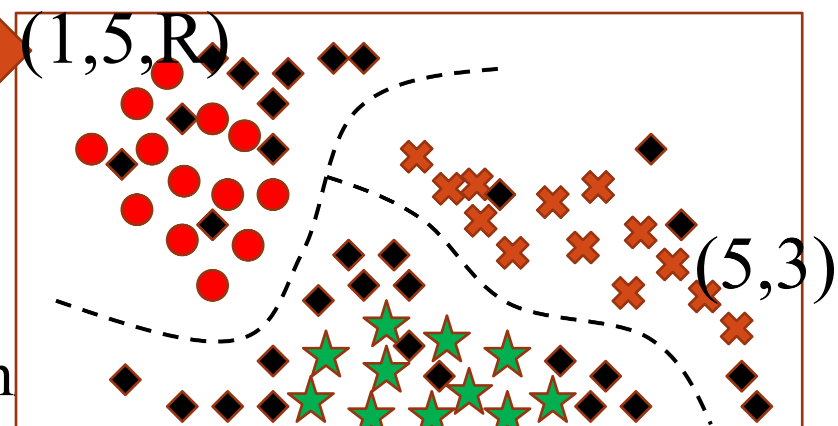
Association: frequent concurrenc



3. Semi supervised Learning

More natural learning problems,

human concept learning



4. Reinforcement Learning

Decision making (robot, chess m

Based on feedback/(reward & penalty)

Supervised & Unsupervised Learning

- Supervised learning is a ML task of inferring a function from labeled training data. The training data consist of a set of training examples.
- Each example is a pair consisting of an input object (typically a vector) and the desired output value/class/label
- If you learn the thing before from training data and then applying that knowledge to the test data(for new fruit), This type of learning is called as Supervised Learning.
- Suppose you had a basket and filled it with different kinds of fruits. Your task is to arrange them into groups.

SI.No.	SIZE	COLOR	SHAPE	FRUIT NAME(Label)
1	Big	Red	Rounded	Apple
2	Small	Red	Heart-shaped	Cherry
3	Big	Green	Long curving cylinder	Banana
4	Smal	Green	Round to oval	Grape

Variable Types

Numerical

Discrete (# items, population, count of students etc)

Continuous (ex: height, weight, profit.)

Interval

Examples: calendar dates, temperatures in Celsius or Fahrenheit.

Ratio

Examples: temperature in Kelvin, length, time, counts

Categorical

Nominal categorical (location, caste, gender)

Ordinal categorical (Grade, hotness, coldness etc.)

Properties of Attribute Values

The type of an attribute depends on which of the following properties it possesses: –

- Distinctness: $= \neq$
- Order: $< >$
- Addition: + -
- Multiplication: * /
- Nominal attribute: distinctness
- Ordinal attribute: distinctness & order
- Interval attribute: distinctness, order & addition
- Ratio attribute: all 4 properties

Sample, tuple, features , dimension in data set

Unsupervised Learning:

- Suppose we have a basket filled with some different types of fruits and the task is to arrange them as groups.
- This time, we don't know name of the fruits, the first time you have seen them.
- So, how to arrange/group them?
- We will take a fruit and will arrange them by considering the physical characteristics of that particular fruit.
- In data mining or machine learning, this kind of learning is known as *unsupervised learning*.

Reinforcement Learning

- Policies: what actions should an agent take in a particular situation (environment)
- Utility estimation: how good is a state (□used by policy)

- No supervised output but delayed reward
- Applications:
  - Game playing
  - Robot in a maze
  - Multiple agents, partial observability, ...
  - Three major components can be identified in reinforcement learning functionality: the agent, the environment, and the actions.

Why Learn?

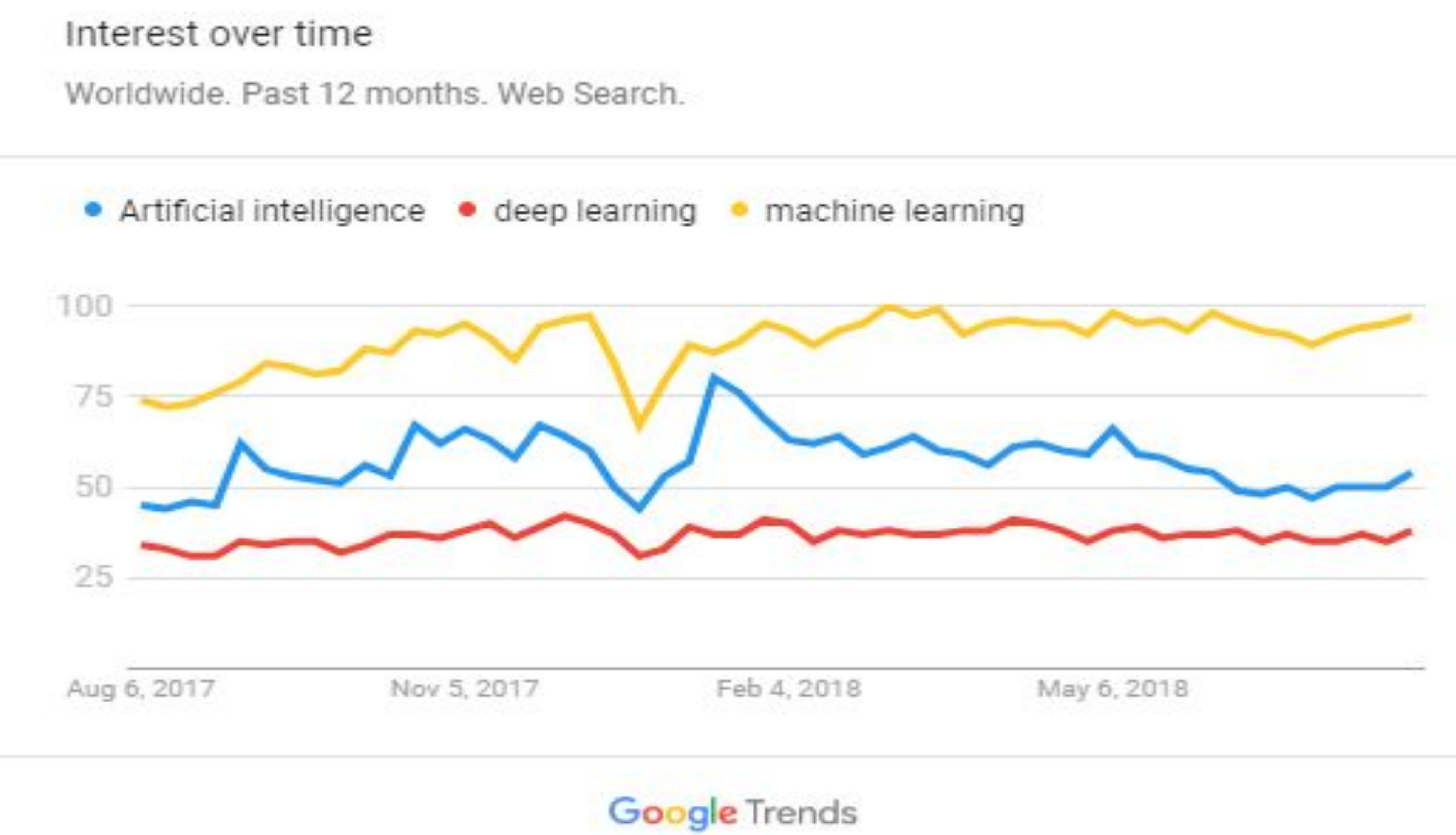
- Learning is used when:
  - Human expertise does not exist (navigating on Mars),
  - Humans are unable to explain their expertise (speech recognition, finger print/retina Recognition)
  - Solution changes in time (routing on a computer network)
  - Data is abundant (data warehouses, data marts); knowledge is expensive and scarce.

Machine learning is preferred approach to

- Speech recognition, Natural language processing
- Computer vision
- Medical outcomes analysis
- Robot control
- Computational biology

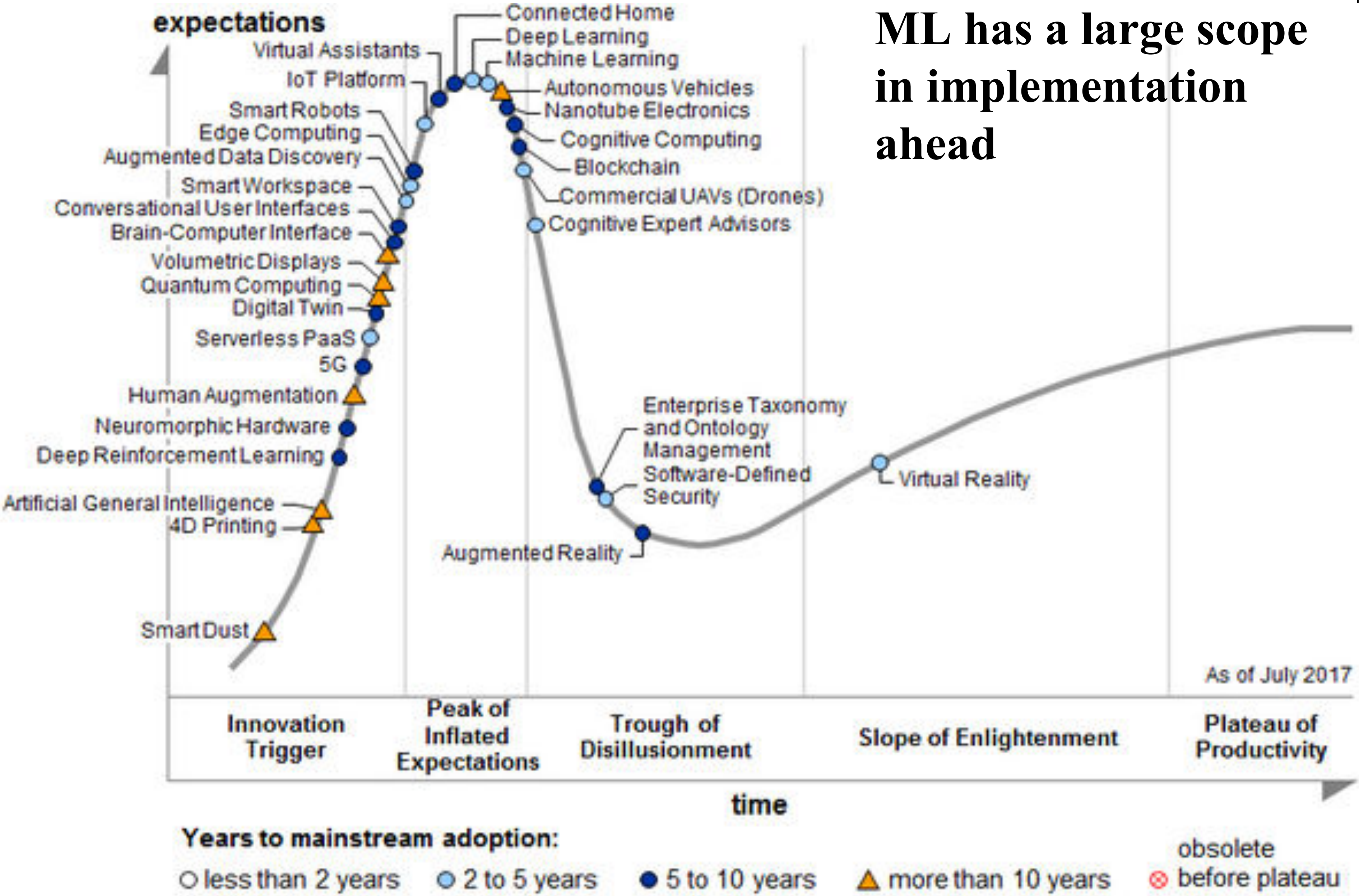
- “A year spent in artificial intelligence is enough to make one believe in God.” —Alan Perlis

“AI technologies will be the most disruptive (innovative or groundbreaking) class of technologies over the next 10 years due to radical (extreme, innovative) computational power, near-endless amounts of data and unprecedented advances in deep neural networks,” **says Mike J. Walker, research director for Gartner.**



Gartner report -2017

ML has a large scope in implementation ahead



Query??